

Conceptual Writing Sample

Disclaimer: I adapted this sample from work I created for a previous employer. I removed all proprietary, confidential, and identifying information, and I rewrote portions of the content to ensure it is neutral and appropriate for public use. The original material belongs to that company. I provide this sample solely to demonstrate my writing ability.

All company names, product names, and scenarios in this sample are fictional and used solely for demonstration purposes.

This conceptual sample demonstrates my ability to explain high-level product concepts to non-technical audiences.

CipherFlow service

The CipherFlow service in GivingBridge protects sensitive configuration values across the platform. SilverTurnip Technologies installs and maintains the required certificate or thumbprint during installation, upgrades, and data migrations. The service ensures that GivingBridge stores confidential settings in a secure and encrypted format.

Security requirements

If you create a security pattern that accesses configurations, you must add the read privilege for the entity. This entity appears on the **Custom** tab in the **Pattern** form. When you upgrade to a new release of GivingBridge, the upgrade process automatically adds the required read privilege to all applicable entities.

Where does GivingBridge store encryption details?

GivingBridge stores encryption information in following locations:

- **Certificate Information** - The entity contains a field named **Cert Name**, which stores the certificate name or thumbprint. This field supports up to 500 characters. SilverTurnip Technologies installs and maintains this value. Unless documentation states otherwise, your organization does not need to install, update, or maintain it.
- **Encrypted Configuration Values** - Each encrypted configuration value appears in a field named **Special** on the **Configuration** page. You can enter a new value in this field. After you save the value, GivingBridge replaces the text with `{{encrypted}}`.

GivingBridge stores the actual encrypted value in a separate field that supports multiple lines and up to 25,000 characters. The service maintains this value automatically.

How does the service work?

The service includes the following capabilities:

- It encrypts configuration values when required. If the certificate name or thumbprint is incorrect, GivingBridge displays an error during the encryption process.
- When SilverTurnip Technologies updates or replaces a certificate, GivingBridge updates the **CertificateName** field. After GivingBridge validates the new certificate, it locates all configuration settings that the previous certificate encrypted, decrypts them with the original certificate, and re-encrypts them with the new certificate.